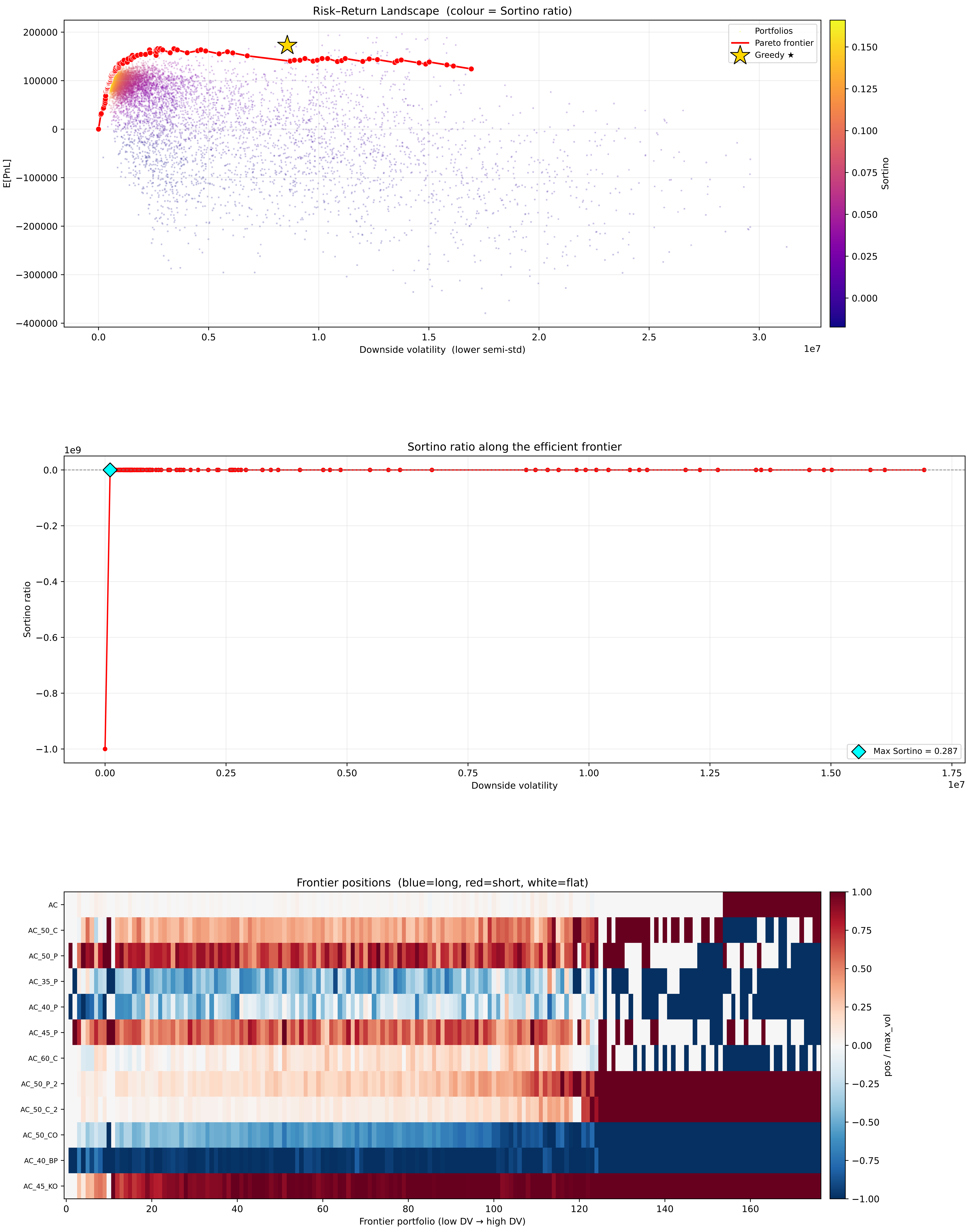
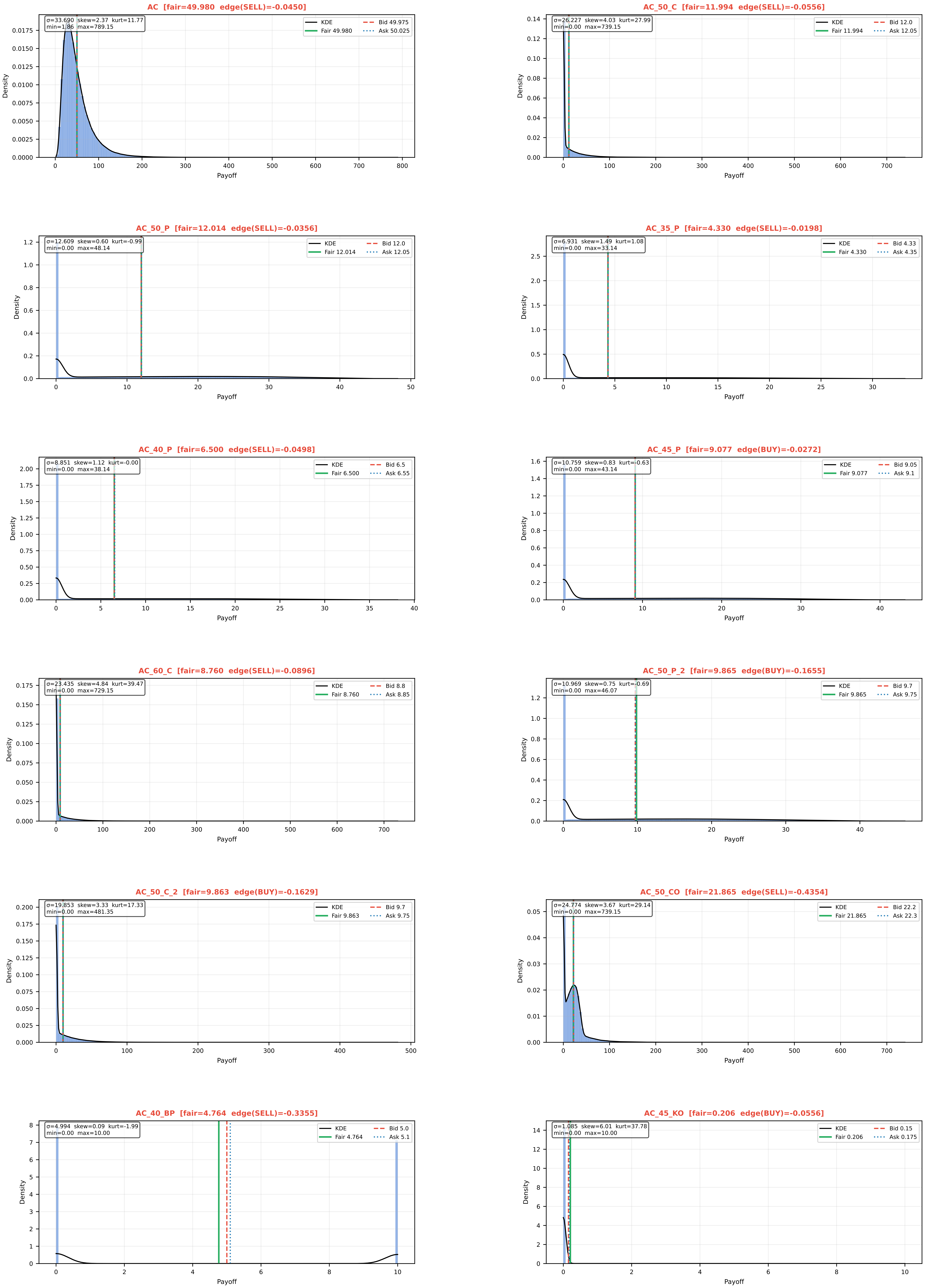
AETHER_CRYSTAL – Efficient Frontier (Global Monte Carlo v4)



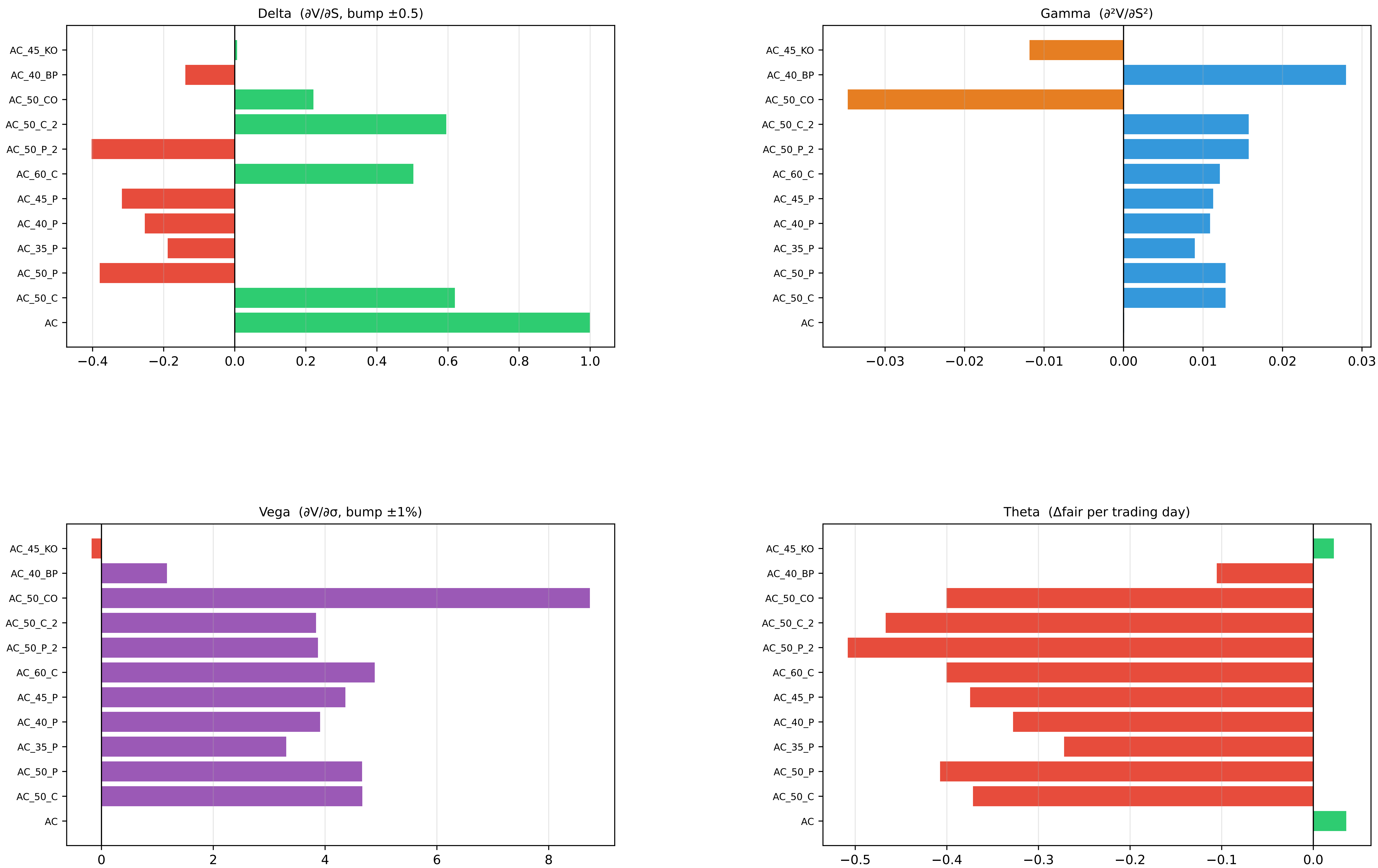
Efficient Frontier – Full Position Table (sorted by E[PnL] desc)

Rank	E[PnL]	DwnVol	Sortino	AC	AC_50_C	AC_50_P	AC_35_P	AC_40_P	AC_45_P	AC_60_C	AC_50_P_2	AC_50_C_2	AC_50_CO	AC_40_BP	AC_45_KO
1	165,813.9	2,685,846.1	0.062	0	0	0	-50	0	0	0	50	50	-50	-50	500
2	165,805.3	2,809,762.7	0.059	0	0	0	0	-50	0	0	50	50	-50	-50	500
3	165,781.4	3,425,655.4	0.048	0	0	0	-50	-50	0	0	50	50	-50	-50	500
4	163,616.8	4,647,630.7	0.035	0	0	-50	-50	-50	0	0	50	50	-50	-50	500
5	163,431.6	2,317,346.4	0.071	0	50	0	0	0	0	-50	50	50	-50	-50	500
6	163,407.7	2,752,126.6	0.059	0	50	0	-50	0	0	-50	50	50	-50	-50	500
7	163,399.1	2,911,437.5	0.056	0	50	0	0	-50	0	-50	50	50	-50	-50	500
8	163,375.2	3,578,009.9	0.046	0	50	0	-50	-50	0	-50	50	50	-50	-50	500
9	162,365.1	2,616,469.9	0.062	0	0	0	-50	-50	50	0	50	50	-50	-50	500
10	161,697.5	4,508,856.6	0.036	0	0	0	-50	-50	-50	0	50	50	-50	-50	500
11	161,210.6	4,866,784.4	0.033	0	50	-50	-50	-50	0	-50	50	50	-50	-50	500
12	159,959.0	2,642,581.2	0.061	0	50	0	-50	-50	50	-50	50	50	-50	-50	500
13	159,533.0	5,855,670.7	0.027	0	0	-50	-50	-50	-50	0	50	50	-50	-50	500
14	158,039.7	2,583,513.2	0.061	0	50	50	-50	-50	0	-50	50	50	-50	-50	500
15	157,497.8	2,345,048.6	0.067	0	50	0	0	0	0	0	50	50	-50	-50	500
16	157,465.2	3,253,357.2	0.048	0	50	0	0	-50	0	0	50	50	-50	-50	500
17	157,441.3	4,027,268.6	0.039	0	50	0	-50	-50	0	0	50	50	-50	-50	500
18	157,126.8	6,095,731.5	0.026	0	50	-50	-50	-50	-50	-50	50	50	-50	-50	500
19	155,276.8	5,474,827.8	0.028	0	50	-50	-50	-50	0	0	50	50	-50	-50	500
20	154,057.7	1,923,402.2	0.080	0	50	0	-50	0	50	0	50	50	-50	-50	500
21	154,049.0	2,132,750.9	0.072	0	50	0	0	-50	50	0	50	50	-50	-50	500
22	152,162.3	1,537,886.7	0.099	0	50	50	0	0	0	0	50	50	-50	-50	500
23	152,138.4	1,767,820.8	0.086	0	50	50	-50	0	0	0	50	50	-50	-50	500
24	152,105.9	2,615,245.1	0.058	0	50	50	-50	-50	0	0	50	50	-50	-50	500
25	151,193.0	6,754,446.5	0.022	0	50	-50	-50	-50	-50	0	50	50	-50	-50	500
26	148,987.7	1,476,709.1	0.101	0	0	0	0	0	50	50	50	50	-50	-50	500
27	148,689.7	1,623,994.5	0.092	0	50	50	-50	-50	50	0	50	50	-50	-50	500
28	147,044.6	1,571,344.9	0.094	0	0	50	-50	0	0	50	50	50	-50	-50	500
29	145,592.6	9,371,521.1	0.016	200	-50	0	0	0	0	-50	50	50	-50	-50	500
30	145,568.6	10,151,781.4	0.014	200	-50	0	-50	0	0	-50	50	50	-50	-50	500
31	145,560.0	10,401,841.1	0.014	200	-50	0	0	-50	0	-50	50	50	-50	-50	500
32	145,536.1	11,201,875.7	0.013	200	-50	0	-50	-50	0	-50	50	50	-50	-50	500
33	145,015.5	1,341,356.6	0.108	1	46	45	-8	-12	-1	-8	50	42	-50	-41	500
34	144,696.1	12,293,498.0	0.012	200	0	0	-50	-50	0	-50	50	50	-50	-50	500
35	143,595.8	1,482,979.4	0.097	0	0	50	-50	-50	50	50	50	50	-50	-50	500
36	143,371.6	12,663,365.5	0.011	200	-50	-50	-50	-50	0	-50	50	50	-50	-50	500
37	142,531.5	13,746,972.4	0.010	200	0	-50	-50	-50	0	-50	50	50	-50	-50	500
38	142,152.3	8,894,508.2	0.016	200	-50	0	-50	0	50	-50	50	50	-50	-50	500
39	142,143.7	9,145,531.3	0.016	200	-50	0	0	-50	50	-50	50	50	-50	-50	500
40	142,119.8	9,930,842.4	0.014	200	-50	0	-50	-50	50	-50	50	50	-50	-50	500
41	142,082.2	1,159,790.6	0.123	5	32	28	-11	-1	20	-3	46	34	-45	-49	487
42	141,279.8	11,038,568.9	0.013	200	0	0	-50	-50	50	-50	50	50	-50	-50	500
43	140,612.3	13,558,820.3	0.010	200	0	0	-50	-50	-50	-50	50	50	-50	-50	500
44	140,233.2	8,702,998.4	0.016	200	-50	50	-50	0	0	-50	50	50	-50	-50	500
45	140,200.7	9,743,646.1	0.014	200	-50	50	-50	-50	0	-50	50	50	-50	-50	500
46	139,602.2	11,996,387.9	0.012	200	-50	0	-50	-50	0	0	50	50	-50	-50	500
47	139,360.7	10,845,761.4	0.013	200	0	50	-50	-50	0	-50	50	50	-50	-50	500
48	138,447.7	15,017,398.4	0.009	200	0	-50	-50	-50	-50	-50	50	50	-50	-50	500
49	137,986.2	1,306,153.0	0.106	-7	35	50	-41	3	-7	-12	33	50	-39	-50	500
50	137,437.8	13,452,320.9	0.010	200	-50	-50	-50	-50	0	0	50	50	-50	-50	500
51	137,119.5	1,052,638.1	0.130	0	50	0	-50	0	50	0	50	0	-50	-50	500
52	136,597.7	14,554,001.1	0.009	200	0	-50	-50	-50	0	0	50	50	-50	-50	500
53	135,475.4	1,105,569.1	0.123	-1	29	39	-1	-20	9	0	31	35	-41	-50	500
54	135,200.3	976,971.5	0.138	0	50	50	-50	0	0	0	50	0	-50	-50	500
55	135,142.8	937,817.8	0.144	14	25	33	-12	4	22	4	39	14	-49	-50	490
56	134,191.5	14,855,119.2	0.009	200	50	-50	-50	-50	0	-50	50	50	-50	-50	500
57	132,513.9	15,815,564.9	0.008	200	0	-50	-50	-50	-50	0	50	50	-50	-50	500
58	132,161.9	936,572.1	0.141	1	28	32	10	-25	28	9	33	22	-47	-47	500
59	130,107.7	16,113,125.2	0.008	200	50	-50	-50	-50	-50	-50	50	50	-50	-50	500
60	129,523.3	932,041.7	0.139	5	19	13	-30	0	31	10	50	13	-38	-50	482
61	129,405.3	919,199.3	0.141	2	47	49	4	-23	22	1	34	18	-50	-50	449
62	127,881.6	882,827.8	0.145	11	22	21	-6	-6	34	8	34	12	-48	-42	500
63	127,165.4	909,372.1	0.140	-1	33	43	22	-40	20	13	27	20	-50	-44	500
64	126,901.1	923,103.6	0.137	9	10	30	-19	-3	25	19	33	20	-41	-50	490
65	126,696.1	795,564.2	0.159	16	19	22	-17	-6	50	2	32	11	-42	-50	500
66	125,893.3	854,491.5	0.147	5	9	30	-9	-27	33	28	36	12	-43	-50	500
67	125,378.2	854,931.8	0.147	11	17	40	-17	-14	37	4	21	23	-41	-50	500
68	124,173.8	16,926,847.4	0.007	200	50	-50	-50	-50	-50	0	50	50	-50	-50	500
69	123,543.5	769,394.3	0.161	-1	30	32	-30	7	22	10	28	14	-44	-50	470
70	121,217.0	757,993.2	0.160	1	30	47	-20	-12	33	10	20	15	-47	-44	500
71	120,484.9	761,579.1	0.158	7	25	37	-14	-15	25	11	26	5	-42	-50	500
72	119,622.3	737,031.8	0.162	6	22	41	-12	-12	22	13	22	10	-43	-48	493
73	119,467.3	723,306.2	0.165	1	29	43	-5	-4	15	10	19	12	-46	-48	456
74	119,193.1	726,623.1	0.164	-1	31	30	-19	14	20	12	18	12	-46	-50	461
75	118,407.4	735,974.8	0.161	-2	27	33	-14	-8	24	18	17	8	-48	-46	474
76	118,146.6	722,882.4	0.163	4	31	44	4	-25	15	2	20	6	-42	-47	500
77	118,089.3	702,599.2	0.168	12	24	41	-3	-10	26	2	14	6	-42	-50	500
78	117,229.1	672,651.4	0.174	7	16	26	-27	0	37	10	23	10	-36	-50	500
79	116,546.6	698,347.2	0.167	-2	37	35	-19	-27	31	-4	21	10	-36	-50	500
80	116,058.3	669,884.2	0.173	3	29	43	-23	-7	23	10	22	6	-39	-50	500
81	115,899.1	651,323.6	0.178	1	21	23	-18	-2	30	9	21	8	-35	-50	500
82	114,576.9	663,054.3	0.173	10	15	35	-12	-21	43	6	14	10	-37	-50	500
83	114,478.6	649,863.5	0.176	5	20	31	-26	-2	32	3	19	10	-34	-50	493
84	113,563.4	616,362.6	0.184	8	24	47	-5	-28	29	4	16	5	-38	-50	500
85	112,586.3	607,479.5	0.185	7	20	37	-20	-10	33	10	20	4	-36	-50	500
86	112,569.9	612,813.0	0.184	4	20	32	-20	-14	38	12	13	5	-38	-50	500
87	110,723.7	604,836.4	0.183	1	18	24	-29	-7	36	13	17	6	-33	-50	500
88	109,507.4	574,700.9	0.191	4	17	37	-37	-4	36	10	11	8	-34	-50	500
89	108,726.4	566,679.1	0.192	3	18	17	-27	4	38	7	17	5	-30	-50	500
90	108,562.3	570,727.7	0.190	0	24	38	-18	-7	17	8	12	3	-34	-49	500
91	107,449.4	564,460.0	0.190	2	14	35	-22	-10	28	15	13	5	-33	-50	490
92	107,212.3	552,332.5	0.194	8	15	41	-12	-16	31	10	13	3	-33	-50	500
93	106,707.9	534,525.3	0.200	3	22	45	-33	0	25	8	9	5	-35	-50	480
94	106,354.7	548,741.6	0.194	7	14	42	-15	-11	31	10	8	5	-33	-50	500
95	105,960.2	562,313.5	0.188	-5	24	28	-31	-6	29	7	12	8	-30	-48	500
96	105,745.5	549,486.2	0.192	3	13	41	-19	-9	17	14	12	3	-31	-50	500
97	105,642.6	531,290.1	0.199	2	19	22	-28	4	35	7	14	3	-29	-50	500
98	105,564.4	557,872.4	0.189	3	20	42	-23	-14	17	5	14	3	-29	-50	500
99	105,265.4	550,427.6	0.191	2	22	48	-13	-34	33	6	7	6	-33	-49	500
100	105,044.8	543,870.7	0.193	7	17	33	-14	-9	31	8	11	3	-33	-47	494
101	104,992.7	539,526.6	0.195	3	18	29	-23	-4	32	8	14	3	-31	-50	473
102	104,832.1	543,912.9	0.193	-3	21	26	-37	2	34	8	13	7	-29	-50	485
103	104,742.4	546,200.9	0.192	4	14	46	-31	-9	21	10	15	6	-28	-50	500
104	104,268.3	540,786.1	0.193	3	21	43	-20	-9	24	6	9	6	-33	-47	484
105	104,145.5	523,813.1	0.199	5	18	33	-17	-11	32	7	9	2	-31	-50	491
106	104,106.6	548,372.8	0.190	3	25	34	-28	-10	33	0	10	6	-31	-49	471
107	103,538.3	519,225.7	0.199	3	19	34	-28	-19	41	5	8	5	-29	-50	500
108	103,301.8	515,959.2	0.200	5	21	41	-18	-18	33	5	10	3	-30	-50	495
109	102,718.4	512,807.6	0.200	0	19	30	-13	-7	24	8	8	2	-28	-50	500
110	102,530.9	512,495.4	0.200	7	15	28	-32	1	50	14	7	4	-36	-50	446
111	102,369.8	536,218.1	0.191	2	16	40	-14	-30	26	7	11	3	-27	-50	500
112	102,336.0	527,537.1	0.194	4	23	50	-11	-15	16	0	10	5	-28	-49	492
113	101,973.0	529,854.8	0.192	4	18	28	-30	-1	40	10	10	4	-34	-41	500
114	101,897.6	526,436.2	0.194	3	15	21	-34	-1	42	10	16	4	-30	-48	453
115	101,862.3	511,300.9	0.199	6	18	41	-14	-14	19	0	11	2	-26	-50	494
116	101,848.1	511,023.4	0.199	4	18	30	-18	-10	30	4	14	3	-26	-50	483
117	101,801.2	506,915.5	0.201	1	19	30	-28	-5	35	12	11	3	-32	-48	468
118	101,785.7	537,416.9	0.189	4	15	50	-30	-16	21	12	9	3	-31	-50	481
119	101,743.8	530,752.3	0.192	0	23	39	-30	-14	27	2	11	3	-28	-47	500
120	101,635.1	526,230.9	0.193	1	22	36	-16	-18	30	9	11	2	-29	-49	495
121	101,288.9	501,666.2	0.202	1	19	37	-32	-10	27	8	12	2	-28	-49	489
122	101,226.5	500,074.8	0.202	1	21	38	-13	-14	23	5	5	2	-28	-50	500
123	101,087.8	525,024.0	0.193	5	15	46	-10	-27	22	2	12	7	-24	-49	500
124	101,028.4	496,874.1	0.203	1	17	31	-23	-10	29	6	9	4	-26	-49	497
125	100,840.5	523,322.5	0.193	-3	23	35	-7	-21	16	7	11	2	-28	-50	473
126	100,723.3	523,661.8	0.192	-1	16	44	-19	-7	11	15	9	5	-29	-49	482
127	100,624.3	491,778.6	0.205	-2	20	21	-28	-4	38	10	10	1	-27	-49	500
128	100,573.3	523,285.9	0.192	-1	20	22	-29	-2	32	8	15	6	-29	-46	454
129	99,999.0	481,800.8	0.208	-1	25	26	-32	-3	42	5	7	1	-29	-50	478
130	99,759.8	477,817.6	0.209	1	18	32	-32	-6	28	6	10	2	-25	-49	500
131	99,597.1	518,731.9	0.192	10	18	42	-20	-1	23	4	8	3	-31	-44	490
132	99,588.4	519,656.2	0.192	0	17	24	-22	-8	24	7	15	4	-25	-45	490
133	99,496.9	522,310.0	0.190	5	21	38	-14	-30	36	4	9	2	-30	-47	480
134	98,650.6	524,104.6	0.188	3	12	35	-45	0	27	10	10	6	-25	-50	472
135	98,618.7	520,456.6	0.189	3	16	33	-21	-22	33	7	9	4	-27	-45	500
136	97,572.8	473,834.7	0.206	3	13	26	-34	3	34	8	6	3	-26	-50	471
137	97,275.3	471,971.1	0.206	-1	23	44	-11	-35	24	0	8	3	-23	-49	500
138	97,273.8	467,651.4	0.208	0	23	34	-11	-29	30	0	8	1	-23	-49	500
139	95,858.8	464,574.9	0.206	3	21	42	-21	-16	25	1	7	4	-26	-50	446
140	95,836.0	459,747.3	0.208	0	18	40	-26	-14	26	6	6	3	-24	-49	486
141	95,696.3	457,652.7	0.209	5	19	44	-26	-12	23	2	10	1	-25	-50	458
142	94,449.7	456,317.6	0.207	2	17	33	-33	-3	25	5	9	2	-25	-47	457
143	93,964.8	456,284.2	0.206	6	15	44	-18	-30	33	4	10	2	-24	-50	454
144	93,685.7	455,033.4	0.206	1	20	34	-22	-5	25	7	7	2	-28	-46	442
145	93,606.0	446,971.7	0.209	4	20	43	-15	-24	26	-1	8	2	-23	-49	463
146	93,456.0	446,208.9	0.209	2	22	43	-23	-31	34	-1	6	3	-24	-49	461
147	93,167.3	443,858.0	0.210	-1	20	32	-16	-19	27	3	9	2	-22	-50	450
148	92,774.6	439,876.7	0.211	6	13	38	-33	-21	40	1	7	5	-21	-50	460
149	91,859.8	434,407.2	0.211	2	16	41	-34	-7	24	0	2	4	-19	-48	498
150	90,849.6	434,033.8	0.209	0	22	40	-24	-17	26	4	6	1	-26	-47	434
151	90,801.4	432,823.9	0.210	3	15	46	-25	-20	19	2	7	3	-20	-48	468
152	90,604.5	431,008.9	0.210	-1	15	22	-30	-3	29	5	9	4	-20	-46	454
153	89,953.4	429,130.8	0.210	5	17	41	-22	-17	24	-3	6	3	-21	-47	454
154	89,947.0	414,411.2	0.217	1	21	37	-27	-11	25	0	6	1	-22	-48	444
155	86,871.9	413,440.2	0.210	1	19	40	-18	-30	27	1	10	3	-21	-50	388
156	86,648.8	409,252.5	0.212	6	14	40	-20	-8	22	2	4	1	-23	-50	391
157	86,576.3	403,543.6	0.215	1	18	39	-25	-14	26	4	3	1	-23	-48	417
158	85,075.5	395,621.8	0.215	-3	9	29	-39	9	14	7	3	3	-14	-50	462
159	83,594.3	393,685.1	0.212	3	17	42	-16	-24	21	0	6	2	-20	-46	407
160	82,706.7	386,557.0	0.214	0	28	41	-25	-22	33	-5	3	2	-23	-47	385
161	82,353.4	378,748.7	0.217	2	21	39	-34	-14	34	1	5	3	-24	-47	357
162	81,199.7	378,279.2	0.215	10	10	41	-13	-31	31	-6	4	0	-15	-50	416
163	80,041.3	374,458.1	0.214	1	17	32	-13	-30	30	-3	9	2	-15	-49	377
164	79,417.0	371,392.1	0.214	3	19	47	-17	-31	23	0	8	2	-22	-49	320
165	78,209.1	365,731.5	0.214	5	16	41	-19	-31	31	-4	8	1	-17	-47	372
166	77,823.8	333,615.7	0.233	0	0	0	-50	0	50	0	0	0	0	-50	500
167	68,523.6	324,705.6	0.211	0	50	50	-50	0	50	0	0	0	-50	-50	0
168	62,068.4	311,383.2	0.199	5	-2	50	-12	-50	29	10	2	5	-13	-50	239
169	58,554.8	301,949.4	0.194	9	1	40	-14	-33	28	7	4	-1	-13	-41	273
170	54,316.4	281,439.7	0.193	4	16	45	-6	-39	27	-8	3	3	-14	-46	159
171	54,131.0	298,700.1	0.181	13	-14	17	-23	-13	38	10	7	6	-10	-35	256
172	51,053.1	272,217.5	0.188	-1	28	42	8	-42	17	-8	3	2	-19	-34	175
173	43,580.8	230,361.9	0.189	-1	12	50	-2	-50	8	-7	6	7	-6	-50	53
174	42,246.4	212,339.4	0.199	13	-2	28	4	-42	38	1	2	0	-11	-35	148
175	31,884.9	125,112.4	0.255	0	0	0	-50	0	50	0	0	0	0	-50	0
176	29,957.0	104,240.8	0.287	0	0	50	0	-50	0	0	0	0	0	-50	0
177	0.0	0.0	-1000000000.000	0	0	0	0	0	0	0	0	0	0	0	0

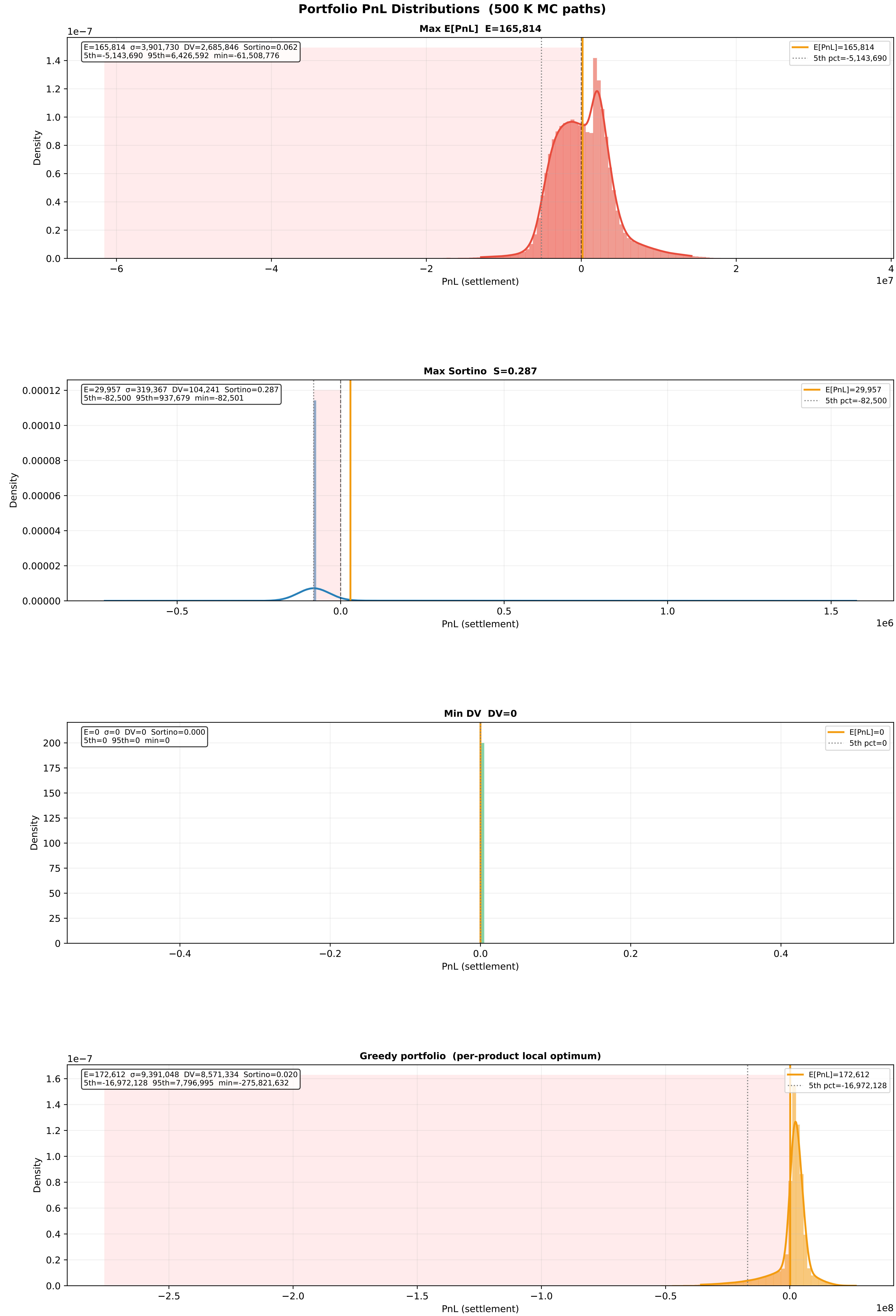
Per-Asset Payoff Distributions (MC simulated)



Per-Asset Greeks (CRN bump-and-reprice, r=0, σ=251%)



Product	Fair	Bid	Ask	Edge(B)	Edge(S)	Delta	Gamma	Vega	Theta/day
AC	49.9492	49.9750	50.0250	-0.0758	+0.0258	+0.9990	+0.00006	+0.0055	+0.03578
AC_50_C	12.0313	12.0000	12.0500	-0.0187	-0.0313	+0.6190	+0.01283	+4.6671	-0.37144
AC_50_P	12.0821	12.0000	12.0500	+0.0321	-0.0821	-0.3800	+0.01282	+4.6619	-0.40722
AC_35_P	4.3618	4.3300	4.3500	+0.0118	-0.0318	-0.1884	+0.00895	+3.3034	-0.27182
AC_40_P	6.5483	6.5000	6.5500	-0.0017	-0.0483	-0.2527	+0.01088	+3.9118	-0.32774
AC_45_P	9.1376	9.0500	9.1000	+0.0376	-0.0876	-0.3170	+0.01126	+4.3627	-0.37452
AC_60_C	8.8094	8.8000	8.8500	-0.0406	-0.0094	+0.5023	+0.01212	+4.8891	-0.40023
AC_50_P_2	9.9282	9.7000	9.7500	+0.1782	-0.2282	-0.4028	+0.01574	+3.8735	-0.50804
AC_50_C_2	9.8269	9.7000	9.7500	+0.0769	-0.1269	+0.5952	+0.01574	+3.8393	-0.46670
AC_50_CO	21.9889	22.2000	22.3000	-0.3111	+0.2111	+0.2214	-0.03468	+8.7351	-0.40036
AC_40_BP	4.7934	5.0000	5.1000	-0.3066	+0.2066	-0.1386	+0.02800	+1.1700	-0.10520
AC_45_KO	0.2024	0.1500	0.1750	+0.0274	-0.0524	+0.0064	-0.01183	-0.1767	+0.02238


Portfolio PnL Distributions (500 K MC paths)
Max E[PnL] E=165,814
E=165,814 σ=3,901,730 DV=2,685,846 Sortino=0.062
5th=-5,143,690 95th=6,426,592 min=-61,508,776
E[PnL]=165,814
5th pct=-5,143,690
Density
PnL (settlement)
1e-7
1e7
Max Sortino S=0.287
E=29,957 σ=319,367 DV=104,241 Sortino=0.287
5th=-82,500 95th=937,679 min=-82,501
E[PnL]=29,957
5th pct=-82,500
Density
PnL (settlement)
1e6
Min DV DV=0
E=0 σ=0 DV=0 Sortino=0.000
5th=0 95th=0 min=0
E[PnL]=0
5th pct=0
Density
PnL (settlement)
Greedy portfolio (per-product local optimum)
E=172,612 σ=9,391,048 DV=8,571,334 Sortino=0.020
5th=-16,972,128 95th=7,796,995 min=-275,821,632
E[PnL]=172,612
5th pct=-16,972,128
Density
PnL (settlement)
1e-7
1e8


Greedy Baseline Analysis
Per-product best edge
AC_45_KO
AC_40_BP
AC_50_CO
AC_50_C_2
AC_50_P_2
AC_60_C
AC_45_P
AC_40_P
AC_35_P
AC_50_P
AC_50_C
AC
−0.4
−0.3
−0.2
−0.1
0.0
Fair − price
Greedy portfolio positions
0
100
200
300
400
500
Position (contracts)
Greedy ★ vs efficient frontier
Frontier
Greedy ★
Max Sortino ◆
E[PnL]
200000
100000
0
−100000
−200000
−300000
0.0
0.5
1.0
1.5
2.0
2.5
3.0
1e7
Downside volatility
Per-product E[PnL] contribution in greedy (total=172,612 DV=8,571,334 Sortino=0.020)
50000
40000
30000
20000
10000
0
AC
AC_50_C
AC_50_P
AC_35_P
AC_40_P
AC_45_P
AC_60_C
AC_50_P_2
AC_50_C_2
AC_50_CO
AC_40_BP
AC_45_KO